BOUNDED GATEWAY PARAMETERS FOR THE ERDŐS–STRAUS CONJECTURE

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ABSTRACT. We present a structural analysis of solutions to the Erdős–Straus conjecture, parameterised by a single integer A. A *gateway decomposition* produces an explicit representation $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ from any divisor $d \mid \left(\frac{p+A}{4}\right)^2$ in a specific residue class modulo A. For $p \equiv 1 \pmod 4$ let $A(p)$ denote the least prime $A \equiv 3 \pmod 4$ whose gateway succeeds, so that $A(p) \geq 3$ always. Three propositions handle all primes outside a residual class — $p \equiv 1 \pmod{24}$, p a quadratic residue mod 7, and every prime factor of $(p+3)/4$ congruent to $1 \pmod 3$ — which requires a computational search over A.

We prove unconditionally that this residual class is negligible:

$$\#\{p \leq X : p \equiv 1 \pmod 4,\ A(p) > 3\} \ \ll\ X(\log X)^{-3/2},$$

so $A(p) = 3$ for a set of primes of density one, and the residual class has relative density $O\big((\log X)^{-1/2}\big)$ among the primes. The value 3 is optimal, since $A(p) \geq 3$ by definition. More generally, for each fixed T, $\#\{p \leq X : p \equiv 1 \pmod 4,\ A(p) > T\} \ll_T X(\log X)^{-1-k(T)/2}$, where $k(T)$ counts the primes $A \leq T$ with $A \equiv 3 \pmod 4$; the case $T = 7$ gives $O(X(\log X)^{-2})$. The proofs are elementary Selberg upper-bound sieves applied to the integers, with "$4n-3$ has no small prime factor" used as a majorant for the primality of $4n-3$; no equidistribution theorem (Bombieri–Vinogradov, the large sieve, GRH) is used anywhere. An earlier version of this paper asserted the density-zero conclusion from a Landau-type count of *integers*; that inference was invalid, and these theorems are what establish the conclusion.

Verification to 10^{11} shows that 32 prime values of A, all congruent to $3 \pmod 4$ and all at most 359, suffice for every prime $p \equiv 1 \pmod 4$ (the value $A = 3$ together with 31 distinct gateway values); primes $p \equiv 3 \pmod 4$ are classical, with $A = 1$. Among the 88 million residual primes, 68.9% are solved at $A = 7$ and 91.0% by $A = 11$. We observe a *slow-growth phenomenon* in the gateway parameter: the maximum A required is 79 at 10^6, 167 at 10^7, 239 at 10^8 and 10^9, and 359 at 10^{10}, unchanged through 10^{11}. This flatness is consistent with an absolute bound but, as Section 7 discusses, does not by itself distinguish a bounded maximum from unbounded growth that is merely slow; an exploratory search beyond 10^{11} has since found two further records, $A = 367$ and $A = 479$, before 1.3×10^{12}. We conjecture that $\max A$ is absolutely bounded and show that this would imply the full conjecture. The observation connects to divisor equidistribution in residue classes for structured quadratic integers.

1. Introduction

The *Erdős–Straus conjecture*, posed by Paul Erdős in 1948, asserts that for every integer $n \geq 2$, the equation

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1}$$

has a solution in positive integers x, y, z.

Prior work. The conjecture has been verified computationally to 10^{17} by Salez [9], who introduced an efficient modular sieve that reduces the set of primes requiring direct verification, and extended to 10^{18} by Mihnea and Bogdan [7]. Earlier verification to 10^{14} is due to Swett [13]. Elsholtz and Tao [3] showed that the average number of solutions satisfies $f(p) \asymp (\log p)^3$ for primes, and Tao [14] observed that quadratic reciprocity presents a fundamental obstruction to resolving the conjecture via covering systems alone.

Algebraic results of Schinzel [10], Sierpiński [12], Mordell [8], and Webb [17], among others, provide explicit decompositions for every residue class modulo 840 except the *Mordell subset* $\{1, 121, 169, 289, 361, 529\}$—the six squares of units modulo 840—which requires methods beyond modular identities.

Vaughn [16] independently formulated an equivalent approach to the hard cases: for each prime p, one seeks x with $\lceil p/4 \rceil \leq x \leq \lceil p/2 \rceil$ and a divisor d of x^2 satisfying a modular condition. Our gateway parameterisation by A (with $x = (p + A)/4$) is algebraically equivalent to this formulation.

Our contribution. We provide a *structural analysis* of solutions, not a new verification record. Using a gateway decomposition parameterised by a single integer A, we classify how each prime is solved and observe a phenomenon not previously noted in the literature:

> *The maximum value of A required to solve all primes up to N grows extremely slowly, over the range verified in this paper: roughly like a small multiple of $\log N$ across five decades, and unchanged from 10^{10} to 10^{11}. Section 7 shows this flatness does not by itself distinguish an absolute bound from unbounded but extremely slow growth, and an exploratory search past 10^{11} has already found two further records.*

We call this the *bounded-A conjecture* (Conjecture 7.1). If true, it implies the full Erdős–Straus conjecture.

The integrality condition underlying the gateway requires $d \mid N^2$ (where $N = (p+A)/4$), which is strictly weaker than $d \mid N$ and essential for completeness.

Our main computational result is:

Theorem 1.1 (Main Result)**.** *Every prime $p \leq 10^{11}$ admits an explicit gateway decomposition* (Theorem 3.1)*: with $A = 1$ if $p \equiv 3 \pmod 4$; with $A = 3$ or $A = 7$ and an explicitly constructed d if $p \equiv 1 \pmod 4$ lies in one of the algebraic classes of Section 4; and otherwise—for the residual class—with a prime $A \leq 359$, $A \equiv 3 \pmod 4$, and a divisor d of $\left(\frac{p+A}{4}\right)^2$ satisfying $d \equiv -p^2 \cdot 4^{-1} \pmod A$, found by computational search.*

Since (1) for composite n reduces to the prime case [8], Theorem 1.1 implies that (1) holds for all $n \leq 10^{11}$.

Alongside this computational result, Section 5 proves two unconditional theorems about the gateway parameter itself (Theorems 5.4 and 5.6): the residual class left behind by the algebraic propositions is not merely small in the verified range but has relative density zero among the primes, at the rate $O((\log X)^{-1/2})$, and $A(p) = 3$ for a set of primes of density one. These are statements about *almost all* primes and say nothing about the boundedness of $A(p)$ for *every* prime, which remains Conjecture 7.1.

The paper is organised as follows. Section 2 fixes notation and recalls the Type I/Type II solution classification of Elsholtz and Tao. Section 3 states and proves the gateway decomposition, noting its equivalence to the formulation of [16]. Section 4 proves that suitable parameters exist for all primes outside a residual class (over 97% of primes up to any fixed bound in our range), and relates this residual class to—while carefully distinguishing it from—the classical Mordell subset. Section 5 proves the unconditional density-one results quoted in the abstract, which show that the residual class left behind by Section 4 has relative density zero among the primes, at the rate $O((\log X)^{-1/2})$; it also corrects an invalid density argument given in an earlier version of this paper (Remark 5.7). Section 6 presents the computational verification for the remaining primes, including A-value distribution data. Section 7 states the bounded-A conjecture, provides heuristic support from the Elsholtz–Tao density estimates, and discusses the connection to divisor equidistribution.

2. Preliminaries

Throughout, p denotes an odd prime. For odd A with $4 \mid (p+A)$, we write

$$N = \frac{p+A}{4}, \qquad B = pN = \frac{p(p+A)}{4}.$$

Note that $A = 4N - p$, so $\frac{4}{p} - \frac{1}{N} = \frac{A}{B}$. Thus (1) with $x = N$ reduces to decomposing $\frac{A}{B}$ as a sum of two unit fractions:

$$\frac{A}{B} = \frac{1}{y} + \frac{1}{z}. \tag{2}$$

The standard factorisation trick for (2) is: if $(Ay-B)(Az-B) = B^2$, then y and z satisfy (2). Setting $\delta_1 = Ay - B$ and $\delta_2 = Az - B$, we need $\delta_1\delta_2 = B^2$ with $\delta_1 \equiv \delta_2 \equiv -B \pmod{A}$.

We use $v_q(m)$ for the q-adic valuation of m, and $\tau(m)$ for the number of positive divisors of m.

3. The Gateway Decomposition

Theorem 3.1 (Gateway Decomposition)**.** *Let p be a prime, A a positive odd integer with $4 \mid (p+A)$. Set $N = \frac{p+A}{4}$ and $B = pN$. Suppose there exists a positive integer d such that*

(i) *$A \mid (B+d)$, and*
(ii) *$d \mid By$, where $y = (B+d)/A$.*

Then

$$x = N, \qquad y = \frac{B+d}{A}, \qquad z = \frac{By}{d} \tag{3}$$

are positive integers, and $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$.

Proof. Integrality of x: immediate from $4 \mid (p+A)$. Integrality of y: immediate from (i). Integrality of z: immediate from (ii). Positivity: clear since $p, A, d > 0$.

For the identity, observe

$$\frac{1}{y} + \frac{1}{z} = \frac{1}{y} + \frac{d}{By} = \frac{B+d}{By} = \frac{Ay}{By} = \frac{A}{B},$$

using $B + d = Ay$ from (i). Since $A = 4N - p$, we have $\frac{A}{B} = \frac{4N-p}{pN} = \frac{4}{p} - \frac{1}{N}$. Therefore $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$. □

Figure 1 illustrates the structure of the decomposition.

The power of Theorem 3.1 lies in identifying *when* a suitable d exists. The following lemma provides a sufficient condition that governs our computational search.

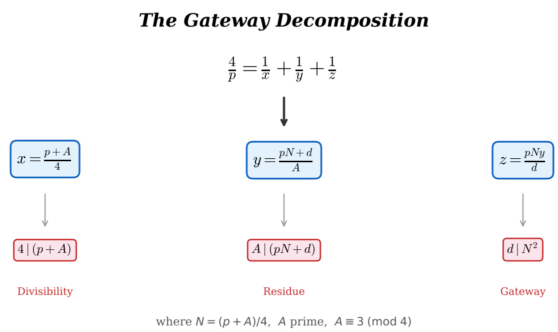

FIGURE 1. Schematic of the gateway decomposition. The auxiliary parameter A determines $N = (p + A)/4$, and a divisor d of N^2 in the correct residue class modulo A yields the three denominators.

Lemma 3.2 (Divisor condition)**.** *In the setting of Theorem 3.1, suppose A is prime, $\gcd(d, pA) = 1$, and $d \mid N^2$. Then condition* (ii) *holds.*

Proof. We must show $d \mid By$ where $y = (B + d)/A$ and $B = pN$. Since $A \mid (B + d)$, write $y = (pN + d)/A$, so $By = pNy = pN(pN + d)/A$. We verify $d \mid pNy$ prime-by-prime.

Let $q^a \| d$. Since $\gcd(d, pA) = 1$, we have $q \neq p$ and $q \neq A$. Write $\alpha = v_q(N)$. From $d \mid N^2$, we get $a \leq 2\alpha$.

- If $\alpha \geq a$: then $q^a \mid N$, so $q^a \mid pNy$. Done.
- If $\alpha < a$: then $\alpha \geq 1$ (since $a \leq 2\alpha$ and $a > \alpha$ gives $\alpha \geq 1$). We have $v_q(pN) = \alpha$ (since $q \neq p$) and $v_q(d) = a > \alpha$, so $v_q(pN + d) = \alpha$. Then $v_q(y) = v_q((pN + d)/A) = \alpha$ (since $q \neq A$). Therefore $v_q(pNy) = \alpha + \alpha = 2\alpha \geq a$. Done.

$\square$

Remark 3.3. The condition $d \mid N^2$ is *strictly weaker* than $d \mid N$, and this distinction is essential. When $\gcd(d, p) > 1$ (e.g., $d = p$ for the identity in Proposition 4.2(b)), the lemma does not apply, but condition (ii) may still hold directly.

Example 3.4. Let $p = 2521$ and $A = 23$. Then $N = (2521 + 23)/4 = 636 = 2^2 \cdot 3 \cdot 53$, and $N^2 = 404{,}496 = 2^4 \cdot 3^2 \cdot 53^2$.

Take $d = 848 = 2^4 \cdot 53$. Then $d \mid N^2$ (since $v_2(d) = 4 \leq 2 \cdot 2 = v_2(N^2)$ and $v_{53}(d) = 1 \leq 2 = v_{53}(N^2)$), but $d \nmid N$ (since $v_2(d) = 4 > 2 = v_2(N)$).

Check (i): $B = 2521 \cdot 636 = 1{,}603{,}356$. $B + d = 1{,}604{,}204 = 23 \cdot 69{,}748$. ✓

So $x = 636$, $y = 69{,}748$, $z = 1{,}603{,}356 \cdot 69{,}748/848 = 131{,}876{,}031$.
Verify: $\frac{4}{2521} = \frac{1}{636} + \frac{1}{69748} + \frac{1}{131876031}$. ✓
This solution *cannot* be found under the condition $d \mid N$.

Remark 3.5 (Equivalence to condition (i)). Since $B = pN$ and $A = 4N - p$, we have $B \equiv pN \pmod{A}$. Now $p \equiv 4N - A \equiv 4N \pmod{A}$, so $B \equiv 4N^2 \pmod{A}$. Thus condition (i) becomes

$$d \equiv -4N^2 \pmod{A},$$

or equivalently (since $\gcd(4, A) = 1$), $d \equiv -p^2 \cdot 4^{-1} \pmod{A}$ (using $p \equiv 4N \pmod{A}$ gives $p^2 \equiv 16N^2 \pmod{A}$, so $-p^2 \cdot 4^{-1} \equiv -4N^2 \pmod{A}$).

Remark 3.6 (Relation to prior formulations). Vaughn [16] independently formulated the Erdős–Straus problem as: for prime p, find x with $\lceil p/4 \rceil \le x \le \lceil p/2 \rceil$ and a divisor $d \mid x^2$ satisfying $d \equiv -px \pmod{4x - p}$. Setting $A = 4x - p$ recovers our parameterisation: $x = N = (p + A)/4$, $4x - p = A$, and the divisor condition matches (i) via Remark 3.5. The two formulations are thus algebraically equivalent. Our contribution is not the identity itself, but the observation that the parameter A stays remarkably small—the bounded-A phenomenon (Section 7).

4. Existence Results

We show that a suitable d exists for specific classes of primes, covering all but a residual class (Remark 4.4), which Section 5 proves has relative density zero among the primes; at 10^6 the three propositions cover 97.1% of primes, rising towards 100% with the bound.

Throughout this section we use the observation that for $A = 3$ and $p \equiv 1 \pmod{4}$, $p \neq 3$: from $4N = p + 3$ we get $N \equiv p \pmod{3}$, hence

$$B = pN \equiv p^2 \equiv 1 \pmod{3}, \tag{4}$$

so condition (i) asks for $d \equiv -1 \equiv 2 \pmod{3}$.

Proposition 4.1. *Every prime $p \equiv 3 \pmod{4}$ satisfies the Erdős–Straus conjecture.*

Proof. This is classical. Set $A = 1$, $N = (p + 1)/4$ (an integer since $p \equiv 3 \pmod{4}$), and $B = pN$. Then $\frac{4}{p} - \frac{1}{N} = \frac{1}{B}$, and $\frac{1}{B} = \frac{1}{B+1} + \frac{1}{B(B+1)}$. So

$$\frac{4}{p} = \frac{1}{(p+1)/4} + \frac{1}{p(p+1)/4 + 1} + \frac{1}{p(p+1)(p(p+1)/4+1)/4}. \qquad \square$$

Proposition 4.2. *Let $p \equiv 1 \pmod{4}$ be prime. Set $A = 3$, $N = (p + 3)/4$, $B = pN$.*

(a) *If* $p \equiv 5 \pmod 8$*, then* N *is even and* $d = 2$ *gives a valid gateway decomposition.*
(b) *If* $p \equiv 17 \pmod{24}$*, then* $d = p$ *gives a valid gateway decomposition.*
(c) *If* $p \equiv 1 \pmod{24}$ *and* N *has a prime factor* $q \equiv 2 \pmod 3$*, then* $d = q$ *gives a valid gateway decomposition.*

Proof. In all cases, $A = 3$, and $4 \mid (p+3)$ since $p \equiv 1 \pmod 4$.

(a): $p \equiv 5 \pmod 8$ gives $N = (p+3)/4$ even, so $2 \mid N$ and hence $2 \mid N^2$. Condition (i) holds since $d = 2$ and, by (4), $B + 2 \equiv 1 + 2 \equiv 0 \pmod 3$ — note this uses only $p \equiv 1 \pmod 4$, $p \neq 3$, and covers both residues of p modulo 3. Lemma 3.2 gives (ii).

(b): Here $d = p$ does not divide N^2 (since $\gcd(p, N) = 1$), so Lemma 3.2 does not apply. Instead we verify (ii) directly. We have $p \equiv 17 \pmod{24}$, so $p \equiv 2 \pmod 3$ and $6 \mid (p+1)$.

Condition (i): $B + p = pN + p = p(N+1) = p(p+7)/4$. Since $p \equiv 17 \pmod{24}$ implies $p \equiv 2 \pmod 3$, we have $p + 7 \equiv 0 \pmod 3$, so $3 \mid (p+7)$ and hence $3 \mid p(p+7)/4 = B + p$. ✓

Condition (ii): $y = (B+p)/3 = p(p+7)/12$ and $By = pN \cdot p(p+7)/12 = p^2(p+3)(p+7)/48$, so $z = By/p = p(p+3)(p+7)/48$. Integrality of z: $p \equiv 17 \pmod{24}$ gives $24 \mid (p+7)$ and $4 \mid (p+3)$, so $48 \mid (p+3)(p+7)$. ✓

(c): q is a prime factor of $N = (p+3)/4$ with $q \equiv 2 \pmod 3$. Since $q \mid N$, we have $q \mid N^2$ and $\gcd(q, 3p) = 1$ (as $q \neq 3$ since $N \equiv 1 \pmod 3$ for $p \equiv 1 \pmod{24}$, and $q \neq p$ since $q \mid N$ and $\gcd(p, N) = 1$). For (i): $B + q = pN + q$. Since $p \equiv 1 \pmod 3$ and $N \equiv 1 \pmod 3$ (from $p \equiv 1 \pmod{24}$), $B \equiv 1 \pmod 3$. Since $q \equiv 2 \pmod 3$, $B + q \equiv 0 \pmod 3$. ✓

Lemma 3.2 gives (ii). □

Proposition 4.3. *Let* $p \equiv 1 \pmod{24}$ *be a prime with every prime factor of* $(p+3)/4$ *congruent to* $1 \pmod 3$ ("Case B")*. If* p *is a quadratic non-residue* $\pmod 7$ ($p \equiv 3, 5,$ *or* $6 \pmod 7$)*, then the gateway decomposition with* $A = 7$ *yields a valid representation.*

Proof. Set $A = 7$, $N = (p+7)/4$, $B = pN$. Since $p \equiv 1 \pmod{24}$ implies $p \equiv 1 \pmod 8$, we have $8 \mid (p+7)$, so $v_2(N) \geq 1$.

We apply Theorem 3.1 directly (not via Lemma 3.2, since our d will not be coprime to p). First, $B \equiv 2p^2 \pmod 7$ (using $4^{-1} \equiv 2 \pmod 7$), and both $2 \equiv 3^2$ and p^2 are QRs mod 7, so B is a QR and $-B$ is a NQR mod 7. Take $d = 2^k p$ with $k \in \{0, 1, 2\}$ chosen so that $2^k p \equiv -B \pmod 7$. This is always possible: $\{2^0, 2^1, 2^2\} \equiv \{1, 2, 4\} \pmod 7$ are the three QRs, and p is a NQR mod 7 by hypothesis, so $\{p, 2p, 4p\}$ runs through all three NQR classes mod 7.

Condition (i) holds by choice of k, so $y = (B+d)/7 = p(N+2^k)/7 \in \mathbb{Z}$.

For condition (ii), we verify $d = 2^k p \mid By$ directly:

- $p \mid By$: $By = pN \cdot p(N+2^k)/7$, so $v_p(By) \geq 2 > 1 = v_p(d)$.
- $2^k \mid By$: $v_2(B) = v_2(N) \geq 1$ and $v_2(y) = v_2(N+2^k)$. For $k = 0$: trivial. For $k = 1$: $v_2(N) \geq 1$ alone suffices. For $k = 2$: if $v_2(N) = 1$, then $N \equiv 2 \pmod 4$ so $v_2(N+4) = 1$ and $v_2(By) = 1+1 = 2$; if $v_2(N) \geq 2$, then $v_2(By) \geq 2$ directly from $v_2(B) \geq 2$.

In every case $v_2(By) \geq k$ and $v_p(By) \geq 1$, so $d \mid By$.✓

Theorem 3.1 now yields the explicit decomposition

$$x = \frac{p+7}{4}, \qquad y = \frac{p(p+7+2^{k+2})}{28}, \qquad z = \frac{By}{d}.$$

□

Remark 4.4. Propositions 4.1–4.3 together cover all primes except those satisfying simultaneously: $p \equiv 1 \pmod{24}$, every prime factor of $(p+3)/4$ is $\equiv 1 \pmod 3$ (Case B), and $p \equiv 1, 2,$ or $4 \pmod 7$. We call these *Case B QR7* primes. At 10^6 they constitute 2.89% of all primes (2,269 of 78,498), and their relative density decreases with the bound (2.38% at 10^9, 2.14% at 10^{11}). That this relative density in fact tends to zero is Corollary 5.5 below; the argument offered for it in an earlier version of this paper—a Landau-type count of the *integers* all of whose prime factors are $\equiv 1 \pmod 3$—does not support the conclusion, and is withdrawn. Remark 5.7 explains precisely why. For these primes we apply Theorem 3.1 with Lemma 3.2 via a computational search over prime values of A.

The residual class is related to, but *not* identical with, the classical *Mordell subset* $\{p : p \bmod 840 \in \{1, 121, 169, 289, 361, 529\}\}$ (the squares of units modulo 840, equivalently $p \equiv 1 \pmod{24}$ with p a quadratic residue modulo both 5 and 7), which is the residual class of the classical system of modular identities. Both sets share the congruence backbone $p \equiv 1 \pmod{24}$ and $\left(\frac{p}{7}\right) = 1$, but they differ in two ways. First, our propositions make no use of the modulus 5: within the gateway family, the constraint $4 \mid (p+A)$ forces $A \equiv 3 \pmod 4$ for $p \equiv 1 \pmod 4$, which excludes $A = 5$, so the classical mod-5 identities do not arise. Thus $p = 193$ (a quadratic non-residue mod 5, hence outside the Mordell subset and covered by a classical mod-5 identity) is nevertheless a Case B QR7 prime and goes to our computational search. Second, the Case-B condition is multiplicative rather than a congruence condition: $p = 3361 \equiv 1 \pmod{840}$ lies in the Mordell subset, yet $(p+3)/4 = 841 = 29^2$ with $29 \equiv 2 \pmod 3$ makes it Case A, so Proposition 4.2(c) handles it algebraically. No system of congruence conditions can capture the residual class exactly. As Tao [14] observed, quadratic reciprocity prevents covering systems from eliminating the Mordell classes; the gateway decomposition circumvents this obstruction by using divisor enumeration on N^2 rather than congruence-based identities.

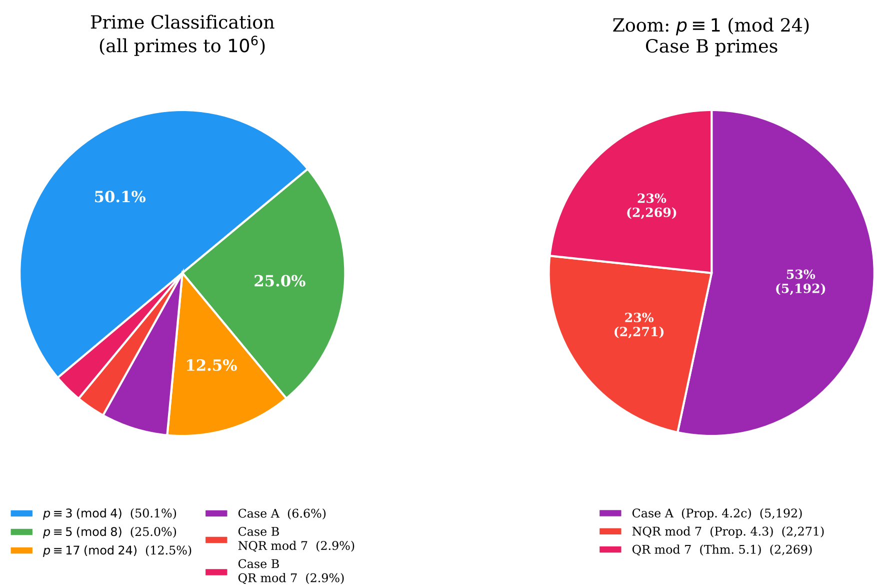

FIGURE 2. Classification of primes to 10^6. *Left:* the four-layer hierarchy; 97.1% of primes are handled by the algebraic results of Propositions 4.1–4.3. *Right:* zoom on the $p \equiv 1 \pmod{24}$ primes, showing the split between Case A, Case B NQR mod 7, and Case B QR mod 7.

5. The Residual Class Has Density Zero

Section 4 leaves a residual class of primes to a computational search. This section proves, unconditionally, that the residual class is negligible: it has relative density zero among the primes, with an explicit rate, and in fact the very first gateway $A = 3$ already succeeds for a set of primes of density one. The proofs are elementary sieve arguments; they use no equidistribution theorem of any kind.

5.1. The gateway function $A(p)$.

Definition 5.1. Let $p \equiv 1 \pmod 4$ be prime. For a prime $A \equiv 3 \pmod 4$ we have $4 \mid (p+A)$, and we say that *gateway A succeeds at p* if there is a divisor d of N_A^2, where $N_A = (p+A)/4$, with

$$d \equiv -B_A \pmod A, \qquad B_A = pN_A,$$

that is, if condition (i) of Theorem 3.1 can be met by a divisor of N_A^2; otherwise gateway A *fails at p*. We write $A(p)$ for the least prime $A \equiv 3 \pmod 4$ whose gateway succeeds at p, and $A(p) = \infty$ if none does. Since 3 is the least prime $\equiv 3 \pmod 4$, we have $A(p) \geq 3$ for every p.

By Remark 3.5 the condition of Definition 5.1 may equivalently be written $d \equiv -4N_A^2 \pmod A$. The next lemma records that a success in the sense of Definition 5.1 always produces an actual decomposition, so that the definition is not weaker than it appears.

Lemma 5.2. *Let $p \equiv 1 \pmod 4$ be prime and $A \equiv 3 \pmod 4$ a prime. Then $\gcd(N_A, A) = 1$. Consequently, if gateway A succeeds at p with divisor d, then $\gcd(d, pA) = 1$, Lemma 3.2 applies, and Theorem 3.1 yields an explicit decomposition $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$.*

Proof. If $A \mid N_A$ then $A \mid 4N_A = p + A$, so $A \mid p$ and $p = A$, impossible since $p \equiv 1$ and $A \equiv 3 \pmod 4$. Hence $\gcd(N_A, A) = 1$, and also $\gcd(N_A, p) = 1$ because $p \mid N_A$ would force $p \mid 4N_A - p = A$. A divisor d of N_A^2 therefore satisfies $\gcd(d, pA) = 1$, and Lemma 3.2 gives condition (ii). □

5.2. Case B is exactly the failure of gateway 3.

The residual class of Section 4 was defined by a congruence condition together with the multiplicative Case-B condition. The following lemma identifies it exactly with the failure of the first gateway, which is what connects Section 4 to the sieve theorems below.

Lemma 5.3. *Let $p \equiv 1 \pmod 4$ be prime. The following are equivalent:*

(i) *gateway 3 fails at p, i.e. $A(p) > 3$;*
(ii) *every prime factor of $N_3 = (p+3)/4$ is $\equiv 1 \pmod 3$;*
(iii) *$p \equiv 1 \pmod{24}$ and every prime factor of $(p+3)/4$ is $\equiv 1 \pmod 3$, i.e. p is a Case B prime in the sense of Proposition 4.3.*

Proof. Write $N = N_3$. By (4) the target residue is $d \equiv 2 \pmod 3$, and by Lemma 5.2 we have $\gcd(N, 3) = 1$, so every prime factor of N is $\equiv 1$ or $2 \pmod 3$.

(ii) $\Rightarrow$ (i): if every prime factor of N is $\equiv 1 \pmod 3$ then so is every divisor of N^2, and none lies in the class $2 \pmod 3$.

(i) $\Rightarrow$ (ii): if some prime $q \mid N$ has $q \equiv 2 \pmod 3$ then $d = q$ divides $N \mid N^2$ and meets the target residue, so gateway 3 succeeds.

(ii) $\Rightarrow$ (iii): the congruence $p \equiv 1 \pmod{24}$ is automatic. Indeed N must be odd, since $2 \equiv 2 \pmod 3$ would be an excluded prime factor; hence $p + 3 = 4N \equiv 4 \pmod 8$ and $p \equiv 1 \pmod 8$. Also $4 \equiv 1 \pmod 3$ gives $N \equiv 4N = p + 3 \equiv p \pmod 3$, and $N \equiv 1 \pmod 3$ (a product of primes $\equiv 1 \pmod 3$), so $p \equiv 1 \pmod 3$. Together, $p \equiv 1 \pmod{24}$.

(iii) $\Rightarrow$ (ii) is trivial. $\square$

Lemma 5.3 is the contrapositive of Proposition 4.2: parts (a), (b) and (c) exhibit, respectively, the divisors $d = 2$, $d = p$ and $d = q$ that make gateway 3 succeed in each of the complementary cases. (Part (b) uses a divisor $d = p \nmid N^2$, so it lies outside Definition 5.1; it is subsumed here because $p \equiv 17 \pmod{24}$ implies $p \equiv 2 \pmod 3$, hence $N_3 \equiv 2 \pmod 3$, hence N_3 has a prime factor $\equiv 2 \pmod 3$ and gateway 3 succeeds in the sense of Definition 5.1 as well.) In particular the Case B QR7 residual class of Remark 4.4 is contained in $\{p : A(p) > 3\}$.

5.3. **The density-one theorems.**

Theorem 5.4.

$$\#\{p \le X : p \equiv 1 \pmod 4,\ A(p) > 3\} \ \ll\ X(\log X)^{-3/2}.$$

The implied constant is absolute; no epsilon, no $\log\log X$ *factor, and no unproved hypothesis is involved.*

Corollary 5.5. *The number of Case B primes up to* X *is* $O\big(X(\log X)^{-3/2}\big)$, *and a fortiori so is the number of Case B QR7 primes. Dividing by* $\pi(X) \sim X/\log X$, *the residual class of Remark 4.4 has relative density* $O\big((\log X)^{-1/2}\big)$ *among the primes; in particular its relative density is zero. Equivalently,* $A(p) = 3$ *for a set of primes* $p \equiv 1 \pmod 4$ *of density one.*

Proof. Immediate from Theorem 5.4 and Lemma 5.3, together with $A(p) \ge 3$ always. $\square$

The value 3 in Theorem 5.4 cannot be lowered: $A(p) \ge 3$ holds by definition, so the theorem says that the least gateway that can ever work is the one that almost always does. The next theorem trades that optimality for a better exponent on a thinner set.

Theorem 5.6. *Fix $T \geq 3$ and let $k(T) = \#\{A \leq T : A$ prime, $A \equiv 3 \pmod 4\}$. Then*

$$\#\{p \leq X : p \equiv 1 \pmod 4,\ A(p) > T\} \ll_T X(\log X)^{-1-k(T)/2}.$$

More generally, for any finite set S of primes $\equiv 3 \pmod 4$, the number of $p \leq X$ with $p \equiv 1 \pmod 4$ for which every gateway $A \in S$ fails is $\ll_S X(\log X)^{-1-|S|/2}$.

The case $T = 3$ ($k = 1$) is Theorem 5.4, and the case $T = 7$ ($k = 2$) gives

$$\#\{p \leq X : p \equiv 1 \pmod 4,\ A(p) > 7\} \ll X(\log X)^{-2},$$

a strictly better exponent on the strictly smaller set $\{A(p) > 7\} \subset \{A(p) > 3\}$. The two statements are logically incomparable: $T = 3$ gives the optimal gateway value on the larger set, $T = 7$ a sharper bound on the smaller one. Note that $\{A(p) > 7\}$ is not the same as the residual class of Remark 4.4 minus the primes covered by Proposition 4.3: that proposition uses a divisor $d = 2^k p$ of By which does not divide N_7^2, so it lies outside Definition 5.1 and a prime it covers may still have $A(p) > 7$. The implied constant in Theorem 5.6 is not uniform in T, and nothing here is claimed as $T \to \infty$.

5.4. **Proof sketches.** We sketch the arguments in enough detail to make the mechanism and its inputs clear, and to let a reader familiar with sieve methods reconstruct them. Full details—in particular the Kneser-theoretic classification of gateway failure underlying Theorem 5.6, and the complete remainder and structure-class bookkeeping—are deferred to a separate paper in preparation.

Theorem 5.4. Reduction. Put $n = N_3 = (p+3)/4$, so that $p = 4n - 3$, and set $x = (X+3)/4$. By Lemma 5.3,

(5)

$$\#\{p \leq X : p \equiv 1 \pmod 4,\ A(p) > 3\} = \#\{n \leq x : 4n{-}3 \text{ prime},\ n \text{ has no prime factor} \equiv 2\ \ (\text{m}$$

This is an exact identity, not an inequality: Lemma 5.3 is an equivalence, and the map $p \mapsto (p+3)/4$ is a bijection between the primes $p \equiv 1 \pmod 4$ and the integers n with $4n - 3$ prime.

The majorant. The right-hand side of (5) is bounded above by sifting the *plain integers* $n \leq x$, not the primes. Fix a sifting level $z \geq 2$ and, at each prime $\ell \leq z$, exclude the residue classes

$$n \equiv 3{\cdot}4^{-1} \pmod \ell \quad (\ell \text{ odd}), \qquad \text{and} \qquad n \equiv 0 \pmod \ell \quad (\ell \equiv 2 \pmod 3)).$$

The first is the class in which $\ell \mid 4n - 3$; it is the standard *rough-number majorant* for the primality of $4n - 3$, since a prime $4n - 3 > z$ has no prime factor $\leq z$ and therefore avoids that class at every $\ell \leq z$.

The second excludes the prime factors $\equiv 2 \pmod 3$ forbidden by (5). Every n counted in (5) with $4n-3 > z$ survives this sieve, so the sifted count exceeds it by at most $O(z)$. Over-counting is harmless: z-rough values of $4n-3$ that are not prime also survive, and an upper-bound sieve is only forbidden to under-count.

This substitution is what makes the argument elementary. The sifted sequence is $\{n \le x\}$, so for a squarefree modulus d the sifted set A_d is a union of $\rho(d)$ residue classes mod d and

$$|A_d| = x\,\frac{\rho(d)}{d} + r(d), \qquad |r(d)| \le \rho(d),$$

by counting integers in an arithmetic progression. *Every remainder term is elementary integer-counting.* No count of primes in arithmetic progressions to a *varying* modulus occurs—the only input of that kind anywhere in the argument is the prime number theorem for the single fixed real character modulo 3, used above to evaluate the density of the sifted primes—and hence no equidistribution theorem—Bombieri–Vinogradov, Barban–Davenport–Halberstam, the large sieve, GRH—is used or needed.

Local data and dimension. The two excluded classes at ℓ coincide only if $3\cdot 4^{-1} \equiv 0 \pmod{\ell}$, that is only for $\ell = 3$, where the second exclusion is not imposed ($3 \not\equiv 2 \bmod 3$). Hence ρ is multiplicative with $\rho(\ell) \le 2$, and $\rho(\ell) < \ell$ at every prime, including $\rho(2) = 1$ and $\rho(3) = 1$. By Mertens' theorem and the prime number theorem for the single fixed real character modulo 3,

$$\sum_{\ell \le z} \frac{\rho(\ell)\log \ell}{\ell} = \log z + \tfrac{1}{2}\log z + O(1) = \tfrac{3}{2}\log z + O(1).$$

The sieve therefore has dimension $\kappa = 3/2$: one full unit from the primality majorant, and one half from the single gateway, the half being the density of the primes $\equiv 2 \pmod 3$.

The sieve. Selberg's Λ^2 upper-bound sieve [11], in the standard form of Halberstam and Richert [5, Ch. 3], with weights supported on $d \le z$, gives

$$S \;\le\; \frac{x}{G(z)} \;+\; \sum_{d \le z^2} 3^{\omega(d)}|r(d)|, \qquad G(z) = \sum_{\substack{d \le z\\ d \mid P(z)}} \mu^2(d)\, g(d),$$

where $P(z)$ is the product of the sifting primes $\ell \le z$ and g is the multiplicative function with $g(\ell) = \rho(\ell)/(\ell - \rho(\ell))$; the factor $3^{\omega(d)}$ counts the ordered pairs (d_1, d_2) with $\operatorname{lcm}(d_1, d_2) = d$. The dimension-$\kappa$ Mertens estimate ([5, Lemma 5.3], valid for arbitrary real $\kappa > 0$)

gives $G(z) \gg (\log z)^{3/2}$. For the remainder, $\rho(d) \le 2^{\omega(d)}$, so by the classical estimate $\sum_{n\le y} m^{\omega(n)} \ll y(\log y)^{m-1}$ [15, Ch. I.3, I.5],

$$\sum_{d\le z^2} 3^{\omega(d)}|r(d)| \;\le\; \sum_{d\le z^2} 6^{\omega(d)} \;\ll\; z^2(\log z)^5.$$

Level. Take $z = x^{1/8}$, so that the weights are supported on $d \le x^{1/8}$ and the remainder is summed over $d \le x^{1/4}$. Then $\log z = \frac{1}{8}\log x$, so the main term is $\ll x(\log x)^{-3/2}$, while the remainder is $\ll x^{1/4}(\log x)^5$: smaller by a power of x, not merely by a power of a logarithm. The boundary loss from $4n-3 \le z$ is $O(x^{1/8})$. Because the remainder is elementary, the level of distribution is nowhere near binding—any fixed positive power of x serves—which is the structural reason no equidistribution input appears. Combining and returning to X proves Theorem 5.4.

Theorem 5.6. Two changes are needed. First, for a general gateway $A \equiv 3 \pmod 4$ there is no exact analogue of Lemma 5.3; the group $(\mathbb{Z}/A)^\times$ has more than one proper subgroup, and gateway failure at N implies only the weaker *necessary* condition that there is a subgroup $K \le (\mathbb{Z}/A)^\times$ of index $c \ge 2$ such that at most $c-2$ distinct prime factors of N have residue outside K. This last condition is the deepest ingredient of the argument and is not elementary: it comes from a classification of gateway failure into a *span* obstruction and a *box* obstruction, obtained by applying Kneser's addition theorem [6] to the set of residues represented by divisors of N^2. We use only the stated consequence; the classification itself is developed in the separate paper in preparation referred to above. For $A = 3$ the only proper subgroup is $\{1\}$, of index 2, and the condition reads "no prime factor outside $\{1\}$", recovering Lemma 5.3 exactly. Second, the gateways are evaluated at different arguments; writing $n = (p+3)/4$ and $h_j = (A_j-3)/4$, one has $N_{A_j} = n + h_j$, so the conditions become conditions on k shifted arguments.

The sieve is then run once for each *structure class* $(K_1, \dots, K_k)$, of which there are $O_T(1)$, and the results are combined by a union bound. For a fixed class the excluded classes are $3 \cdot 4^{-1}$ from the primality majorant and $-h_j$ at each prime ℓ whose residue lies outside K_j; the first never collides with the others except at $\ell = A_j$, where the jth condition is not imposed, and two gateway classes collide exactly when $\ell \mid h_i - h_j$, hence only for the bounded set of primes $\ell \le (T-3)/4$. Finitely many primes change the Euler product by a bounded factor and the exponent not at all. By the prime number theorem in arithmetic progressions to the fixed modulus A_j, the primes outside K_j have density $1 - 1/c_j$, so the dimension of the class is

$$\kappa(K) = 1 + \sum_{j=1}^{k}\Big(1 - \frac{1}{c_j}\Big),$$

again one unit from the primality majorant and $1 - 1/c_j$ per gateway. The classes with some $c_j > 2$ permit up to $c_j - 2$ exceptional prime factors; these are handled by extracting the (bounded) set of exceptional primes and re-running the sieve on the residue class they determine, at a cost of a factor $(\log\log x)^{\sum_j (c_j-2)}$. That cost is affordable, because such a class also saves a genuine power of $\log x$: relative to the all-quadratic-residue class $c_1 = \cdots = c_k = 2$, its bound carries the extra factor

$$(\log x)^{-\sum_j (1/2-1/c_j)}(\log\log x)^{\sum_j (c_j-2)} \longrightarrow 0,$$

since $1/2 - 1/c_j \geq 1/6$ whenever $c_j \geq 3$. The maximum over the $O_T(1)$ classes is therefore attained at the all-quadratic-residue class, which has no exceptional factors at all and dimension $\kappa = 1 + k/2$. The remainder is elementary exactly as before, and z is again a fixed positive power of x. This gives $\ll_T x(\log x)^{-1-k/2}$, and Theorem 5.6 follows. The variant for an arbitrary finite set S is the same argument with the translation "$A(p) > T$ iff every gateway $A \leq T$ fails" replaced by the hypothesis on S directly; nothing else changes.

5.5. **Remarks.**

Remark 5.7 (The earlier density argument, and why it fails). An earlier version of this paper justified the density-zero claim of Remark 4.4 as follows: the Case-B condition forces every prime factor of $(p+3)/4$ into the class 1 (mod 3), and by Landau-type counting [15, Ch. II.5] the integers $n \leq x$ with that property number $\sim c\,x/\sqrt{\log x}$; therefore, it was said, the residual class has relative density zero among the primes.

That inference is invalid. The Landau count is a count of *integers*. Applying it as the reduction (5) requires—to $n = (p+3)/4 \leq (X+3)/4$, not to integers up to X—and dividing by $\pi(X) \sim X/\log X$ gives

$$\frac{c\,(X/4)/\sqrt{\log(X/4)}}{X/\log X} \;\sim\; \frac{c}{4}\sqrt{\log X} \;\longrightarrow\; \infty,$$

so the ratio does not tend to zero; it grows without bound, and the inference fails. Numerically, with $c = 0.3012$, the Landau bound is 2.14×10^4 at $X = 10^6$ against $\pi(X) = 7.85 \times 10^4$, 1.71×10^7 at 10^9 against 5.08×10^7, and 1.54×10^9 at 10^{11} against 4.12×10^9: a bound of the same order as $\pi(X)$, occupying a *rising* fraction of it (0.27, 0.34, 0.37), rather than a vanishing one.

The missing ingredient is that the primality of p must itself be sieved for, and that is what supplies the extra full power of $\log X$; in the language of the sketch above, it is the dimension 1 contributed by the primality majorant, on top of the dimension 1/2 that the Landau count already sees. The conclusion asserted in the earlier version is correct—it is Corollary 5.5—but Theorem 5.4, not the Landau count, is what establishes it.

Remark 5.8 (Sharpness). The exponent $-1 - k(T)/2$ is presumably the true order, and no matching lower bound is claimed or attempted here; every bound in this section is an upper bound. The reason for expecting sharpness is that the all-quadratic-residue structure class, which sets the bound in the proof of Theorem 5.6, is realisable: it is a genuine failure mode of every gateway, not an artefact of the union bound. Numerically, for $T = 3$ the proportion of primes $p \equiv 1 \pmod 4$ with $A(p) > 3$, multiplied by $\sqrt{\log X}$, is constant to within one per cent (at approximately 0.43) over $10^5 \leq X \leq 3 \times 10^8$, which is consistent with $(\log X)^{-1/2}$ being the true relative density rather than merely an upper bound. Since the sieve is one-sided, this remains an observation, not a theorem.

Remark 5.9 (What these theorems do not say). Three limitations deserve to be stated plainly.

(a) A density-one statement permits an infinite exceptional set, of density zero. Theorem 5.4 constrains that set only in size, not in structure, and says nothing about which primes lie in it.
(b) Nothing here bears on whether $A(p)$ is bounded for *every* prime. That is Conjecture 7.1, and it remains open; Theorem 5.6 holds for each fixed T with a constant that is not uniform in T, so it cannot be summed or taken to a limit to yield a bound valid for all primes.
(c) Nothing here bears on the Erdős–Straus conjecture itself, which is in any case verified to 10^{18} [7], far beyond any height at which a density-one statement would first become informative. These are structural results about the gateway parameter, not existence results for (1).

Remark 5.10 (Attribution). The theorems of this section are new. The sieve machinery they use is entirely classical: Selberg's Λ^2 upper-bound sieve [11], the dimension-κ Mertens estimate of Halberstam and Richert [5, Lemma 5.3], the prime number theorem in arithmetic progressions to a fixed modulus, and the divisor-moment estimate $\sum_{n \leq y} m^{\omega(n)} \ll y(\log y)^{m-1}$ [15, Ch. I.3, I.5]. The only novelty in the method is its routing: sifting the integers rather than the primes, so that primality enters as a majorant and every remainder term becomes elementary. Earlier drafts of this material attacked the same statement by sifting the primes and required a divisor-weighted Bombieri–Vinogradov input, which was neither available nor, as it turned out, necessary.

Remark 5.11 (Relation to the divisor literature). The nearest published relative is the work of Banks, Friedlander and Luca [1], who obtain an asymptotic for the count of integers $n \le x$ having *no* divisor $d > 1$ in a fixed residue class $a \pmod q$. Their answer is governed by $H(a)$, the subgroup of $(\mathbb{Z}/q\mathbb{Z})^\times$ of largest order not containing a: writing $\mathcal{P} = [(\mathbb{Z}/q\mathbb{Z})^\times : H(a)]$, the count is $\asymp x(\log\log x)^{\mathcal{P}-2}(\log x)^{-(1-1/\mathcal{P})}$. For $q = A \equiv 3 \pmod 4$ and $a = -4$ the relevant subgroup is $\mathrm{QR}(A)$ and $\mathcal{P} = 2$, giving exponent $1/2$ — the same exponent that governs the Landau count in Remark 5.7, and for the same structural reason.

Their result does not cover the question studied here, on three counts: it concerns divisors of n rather than of n^2; the target class is fixed, whereas the target $-4N^2 \pmod A$ moves with N; and it excludes the divisor $d = 1$, which is admissible in condition (i). Most importantly it is a statement about integers, and Remark 5.7 records why a count over integers does not transfer to the primes. The theorems of this section are for primes, for k gateways simultaneously, and at k distinct shifted arguments; the extra full unit in the exponent $1+k/2$ is exactly the cost of detecting primality, for which [1] has no analogue. Their paper is also explicitly non-uniform in q.

6. Computational Results

For the remaining Case B QR7 primes, we apply the gateway decomposition (Theorem 3.1) with a systematic search over prime values of A.

6.1. **Method.** For each prime p in the Case B QR7 class:

(1) For each prime $A \equiv 3 \pmod 4$ with $4 \mid (p + A)$, in increasing order:
(2) Compute $N = (p + A)/4$ and factorise N.
(3) Enumerate all divisors of N^2 (obtained by doubling the exponents in the prime factorisation of N).
(4) For each divisor d of N^2, check whether $d \equiv -B \pmod A$ where $B = pN$.
(5) If so, compute $y = (B + d)/A$ and $z = By/d$; verify integrality and the identity $\frac{4}{p} = \frac{1}{N} + \frac{1}{y} + \frac{1}{z}$.
(6) Record the first success and move to the next prime.

6.2. **Results.**

Theorem 6.1. *Every prime $p \leq 10^{11}$ admits a gateway decomposition with $A \leq 359$.*

The verified progression by scale (counts and $\max A$ from the corrected pipeline; see Remark 6.2):

Limit	Primes	Case B QR7 primes	Max A
10^6	78,498	2,269	79
10^7	664,579	18,019	167
10^8	5,761,455	146,227	239
10^9	50,847,534	1,212,383	239
10^{10}	455,052,511	10,246,512	359
10^{11}	4,118,054,813	88,088,687	359

Below 10^9, exactly one prime requires $A > 199$: $p = 32{,}349{,}601$, solved by $A = 239$ with $d = 40{,}437{,}300$; the only other prime below 10^9 with $A > 191$ is $p = 597{,}188{,}089$ ($A = 199$). Below 10^{11}, exactly 47 primes require $A \geq 199$ (12 of them below 10^{10}), and only five require $A \geq 251$: $p = 3{,}807{,}728{,}761$ ($A = 359$, $d = 1{,}935$); $p = 31{,}048{,}497{,}481$ ($A = 347$); $p = 6{,}372{,}847{,}849$ ($A = 311$); $p = 26{,}147{,}464{,}729$ ($A = 251$); $p = 94{,}604{,}730{,}049$ ($A = 251$). The largest gateway parameter over the entire range, $A = 359$, is attained below 10^{10} (at $p = 3{,}807{,}728{,}761$) and is not exceeded anywhere in $(10^{10}, 10^{11}]$.

Remark 6.2 (Data integrity). An earlier run of the 10^{10} stage (March 2026) contained an inverted Case A/Case B test: it skipped every residual-class prime as algebraically proven and instead searched the complementary class (Case A $\cap$ QR7), which Proposition 4.2(c) already covers with $A = 3$. All figures previously derived from that run (a residual count of 18,189,229 at 10^{10}, $\max A = 251$, and a 25-value A-dictionary) described the wrong set of primes and have been withdrawn. The corrected pipeline validates against three independent baselines within the same run—the full A-distribution at 10^6, the residual count at 10^7, and the residual count and $\max A$ at 10^9—before extending the range; every solution it reports is verified by direct evaluation of the identity (1). The full referee record accompanies the code repository.

The extension from $d \mid N$ to $d \mid N^2$ is not a cosmetic generalisation: a nontrivial proportion of Case B QR7 solutions use a divisor $d > N$ that would be missed under the stronger condition. Figure 3 shows the distribution of d/N ratios for Case B QR7 solutions.

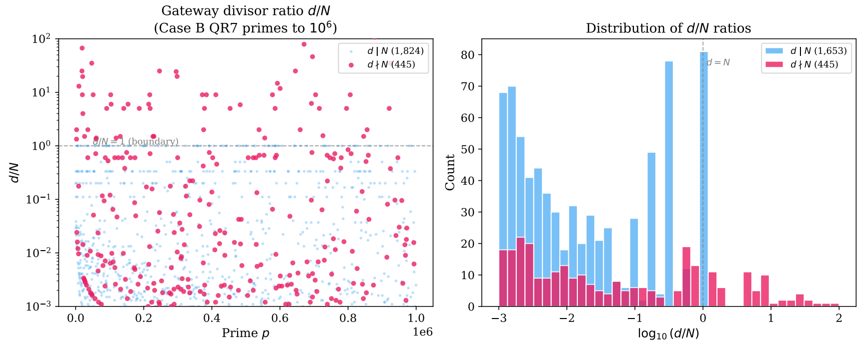

FIGURE 3. Distribution of d/N ratios for Case B QR7 solutions at 10^6. Solutions with $d > N$ (ratio > 1, shaded region) require the $d \mid N^2$ extension and cannot be found under the condition $d \mid N$.

6.3. **Distribution of A values.** Among the 88,088,687 Case B QR7 primes below 10^{11}:

A	Count	% solved	Cumulative
7	60,704,131	68.9%	68.9%
11	19,439,771	22.1%	91.0%
19	4,447,883	5.0%	96.0%
23	2,486,127	2.8%	98.9%
31	688,365	0.8%	99.6%
43–191	322,363	0.37%	99.99995%
≥ 199	47	0.00005%	100%

The extreme sparsity of large-A cases—only 47 primes out of 88 million require $A \geq 199$, and only five require $A \geq 251$—establishes that large-A primes are rare within this range; Section 7 discusses why sparsity of this kind does not, by itself, distinguish a bounded maximum from unbounded but extremely slow growth. Figure 4 shows the distribution at 10^6.

6.4. **Anatomy of the hardest prime below 10^{11}.** The hardest prime below 10^{11}, $p = 3{,}807{,}728{,}761$—the only one in that range requiring $A = 359$—shows why the Case B QR7 class is the computational bottleneck and why the $d \mid N^2$ extension is necessary. (An exploratory search past 10^{11} has since found harder primes; Section 7.)

Why p is Case B.. Here $(p+3)/4 = 951{,}932{,}191 = 7 \cdot 73 \cdot 1{,}862{,}881$, and all three prime factors are $\equiv 1 \pmod 3$. There is no factor $q \equiv 2$

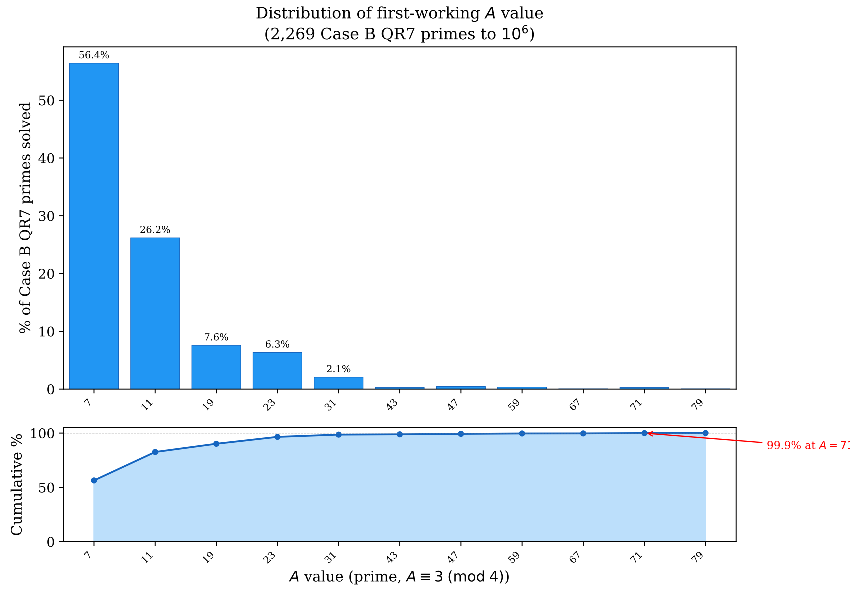

FIGURE 4. Distribution of gateway parameter A for Case B QR7 primes at 10^6. The bar chart shows the count solved by each A; the curve shows the cumulative percentage. A handful of small primes ($A = 7, 11, 19, 23$) handle over 97% of cases.

(mod 3) to supply the shortcut of Proposition 4.2(c), so p falls to the search.

Why every $A < 359$ fails. For each of the 35 candidate primes $A \equiv 3 \pmod 4$ below 359, the target residue $t \equiv -p^2 4^{-1} \pmod A$ must be met by a divisor of N_A^2, where $N_A = (p + A)/4$. The divisor-residue set $\mathcal{R}_A = \{\, d \bmod A : d \mid N_A^2 \,\}$ never contains t: across all 35 values it covers at most 184 of the $A-1$ nonzero residues (at most 83%), because each N_A has only two to six distinct prime factors and its divisors cannot generate enough residues. The obstruction is arithmetic, not congruential.

Why $A = 359$ succeeds. The factorisation

$$N_{359} = \tfrac{p+359}{4} = 951{,}932{,}280 = 2^3 \cdot 3 \cdot 5 \cdot 13 \cdot 23 \cdot 43 \cdot 617$$

has *seven* distinct prime factors—more than any smaller candidate—and its divisor-residue set attains all 358 nonzero residues modulo 359: $|\mathcal{R}_{359}| = 358$. The target $t = 140$ is covered by $d = 1935 = 3^2 \cdot 5 \cdot 43$, giving the explicit solution

$$x = 951{,}932{,}280,$$
$$y = 10{,}096{,}657{,}161{,}783{,}585,$$
$$z = 18{,}913{,}360{,}422{,}476{,}412{,}211{,}734{,}567{,}822{,}280.$$

The $d \mid N^2$, $d \nmid N$ phenomenon. The solving divisor $d = 3^2 \cdot 5 \cdot 43$ divides N_{359}^2 but not N_{359}: $v_3(N_{359}) = 1$ while $v_3(d) = 2$. This is a genuine instance of Lemma 3.2—for this prime the weaker condition $d \mid N^2$ is exactly what is needed.

Isolation. The difficulty is a property of p alone, not its residue class: every Case B QR7 prime within ± 2000 of p is solved by some $A \le 19$. The hardness comes from the factorisation $(p+3)/4 = 7 \cdot 73 \cdot 1{,}862{,}881$, dominated by the large prime 1,862,881.

Why full coverage occurs. A discrete-logarithm analysis (base $g = 13$, a primitive root mod 359) explains $|\mathcal{R}_{359}| = 358$. With $359 - 1 = 2 \cdot 179$ and 179 prime, the index group is $\mathbb{Z}/358\mathbb{Z}$. The seven prime factors of N_{359} split by Legendre symbol into quadratic residues $\{2, 3, 5, 23\}$ (even index, order 179) and primitive roots $\{13, 43, 617\}$ (odd index, order 358). Coverage is forced in four steps:

(1) Since $A \equiv 7 \pmod 8$ and $p \equiv 1 \pmod 8$, $8 \mid (p + A)$, so $v_2(N_{359}) = 3$: the factor 2 is unavoidable.
(2) The residue factors $\{2, 3, 5, 23\}$ generate an arc sumset $E \subset \mathbb{Z}/179\mathbb{Z}$ with $|E| = 147$ and $E \cup (E + 1) = \mathbb{Z}/179\mathbb{Z}$: E is 1-*spanning*.
(3) The primitive root 13 (index 1) shifts the even layer onto the odd layer, reaching all 179 even and 147 odd residues of $\mathbb{Z}/358\mathbb{Z}$.
(4) The primitive root 43 (index 3, $\gcd(3, 179) = 1$) fills the remaining 32 odd residues, completing coverage.

The factor 617 is redundant: six of the seven factors already suffice. Full coverage comes not from primitive roots per se, but from factors whose index generates the correct quotient of $\mathbb{Z}/(A-1)\mathbb{Z}$.

A second mechanism. Below 10^{11}, five primes require $A \ge 251$:

p	A	$(p+3)/4$
3,807,728,761	359	$7 \cdot 73 \cdot 1{,}862{,}881$
31,048,497,481	347	$33{,}457 \cdot 232{,}003$
6,372,847,849	311	$7 \cdot 43 \cdot 1{,}951 \cdot 2{,}713$
26,147,464,729	251	$37 \cdot 1{,}291 \cdot 136{,}849$
94,604,730,049	251	$7 \cdot 37 \cdot 367 \cdot 248{,}821$

The second-largest, $p = 6{,}372{,}847{,}849$ with $A = 311$, reaches full coverage differently. Here $N_{311} = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 632{,}227$, and the single large factor 632,227 has multiplicative order 10 mod 311—not a primitive root—yet its index 217 satisfies $\gcd(217, 310) = 31$, so adjoining it lifts coverage from the even subgroup (155/310) to all 310 residues in one step. Where $A = 359$ needs two primitive roots in sequence, $A = 311$ is completed by a single non-primitive factor of the right index.

7. Discussion

7.1. **The bounded-A conjecture.** The maximum A value grows extremely slowly with scale, over the range formally verified in this paper:

Limit	max A	Increase from previous
10^6	79	—
10^7	167	+88
10^8	239	+72
10^9	239	+0
10^{10}	359	+120
10^{11}	359	+0

Every prime below 10^9 is solved by some $A \leq 239$, and only two primes below 10^9 require $A > 191$. At 10^{10} the maximum rises to 359; it then holds at 359 through 10^{11}, so across that decade $\max A$ does not grow at all.

It is tempting to read this flatness as evidence for Conjecture 7.1, and the original version of this paper did so. That reading does not survive scrutiny. A distributional analysis of the tail of $A(p)$ over this range shows that $\Pr[A(p) \geq T]$ decays as a power of a slowly growing function of $\log X$, and that two candidate models for this decay — one giving an absolutely bounded maximum, the other giving unbounded but extremely slow growth — fit the six data points above *identically*: both predict $79, 167, 239, 239, 359, 359$ exactly, and diverge only far outside the verified range. A single decade of flatness cannot distinguish them; it is what both models predict. The full analysis, including the fitted models and their extrapolations, is available in this repository.[1]

Consistent with this, an exploratory search beyond 10^{11} (same verified pipeline, not independently cross-validated at the same standard as the table above — see the remark below) has since found two further records, at two separate jumps, before reaching 1.212×10^{12} with no further record:

[1]`https://github.com/m4cd4r4/erdos-straus-gateway/blob/main/extended-search/phase3a_growth_law.md`.

p	$\max A$	Increase from previous	
—	359	—	(carried from 10^{11})
235,229,251,009	367	+8	first new record
418,383,886,321	479	+112	second new record
1.212×10^{12} (endpoint)	479	+0	search stopped here

Both record witnesses were independently re-derived by a separate reviewer, and — since — every witness with $A \geq 100$ in the range, 19,540 in total, has been independently re-confirmed the same way (Remark 7.2). The two jumps are of different sizes (+8, then +112); together they total +120 over the interval, comparable to the single $239 \to 359$ jump, but as two separate events rather than one. Either way, the occasional-large-jump pattern from the verified range continues past 10^{11} rather than flattening further. None of this refutes Conjecture 7.1 — 479 is a modest value, and both models above remain live possibilities — but it removes the flatness argument as evidence for it. The conjecture is stated below as an open question, not as one the data supports.

Conjecture 7.1 (Bounded gateway parameter)**.** There exists an absolute constant C such that for every prime $p \equiv 1 \pmod 4$, there is a prime $A \leq C$ with $A \equiv 3 \pmod 4$, $4 \mid (p + A)$, and a divisor d of $\left(\frac{p+A}{4}\right)^2$ satisfying $d \equiv -B \pmod A$, where $B = p(p + A)/4$. Equivalently, $A(p) \leq C$ for all such p, in the notation of Definition 5.1.

The restriction to $p \equiv 1 \pmod 4$ is not a weakening: primes $p \equiv 3 \pmod 4$ are settled unconditionally by Proposition 4.1, and for them $4 \nmid (p + A)$ whenever $A \equiv 3 \pmod 4$, so the displayed condition is vacuous rather than false.

If Conjecture 7.1 holds, the Erdős–Straus conjecture follows, since the conjecture can be verified for all primes up to C by direct computation, and every larger prime admits a gateway with $A \leq C$.

Remark 7.2 (Verification standard for the extended range)*.* The table to 10^{11} is verified to the same standard throughout this paper: the classification of every prime is cross-checked by an independently written implementation sharing no code with the primary pipeline (Section 6). The extension to 1.212×10^{12} uses the same, separately validated search engine. Every witness with $A \geq 100$ in that range — all 19,540 of them, not a sample — has since been independently re-derived: an exhaustive check, for each, that the claimed divisor is genuine and that every smaller candidate gateway fails, with zero discrepancies (`extended-search/tier1_verification.md` in the companion repository). What remains below that standard is the classification of the ordinary primes solved by small A — the vast majority of the range — which has not been independently cross-checked. Closing that gap (matching the 10^{11} table's standard exactly) was costed and deliberately not done: an independent implementation classifies primes at 13–19 thousand per second depending on scale, so the 33.5 billion primes in $[10^{11}, 10^{12}]$ alone would cost roughly 1–3 days of wall time even parallelised (`extended-search/calibrate_tier2_verify.py`). Given that the scientific claim at stake — the value of $\max A$ — is already exhaustively settled by the witness check above, and a systematic bug severe enough to silently misclassify billions of easy primes would be very likely to also corrupt the gateway search already re-verified, the additional confidence was judged not to justify several days of compute. It is reported at this standard deliberately, rather than folded into the 10^{11} table.

A weaker but still significant form: $\max A(N) = O(\log\log p)$. Even $\max A = o(\log p)$ would go beyond current results.

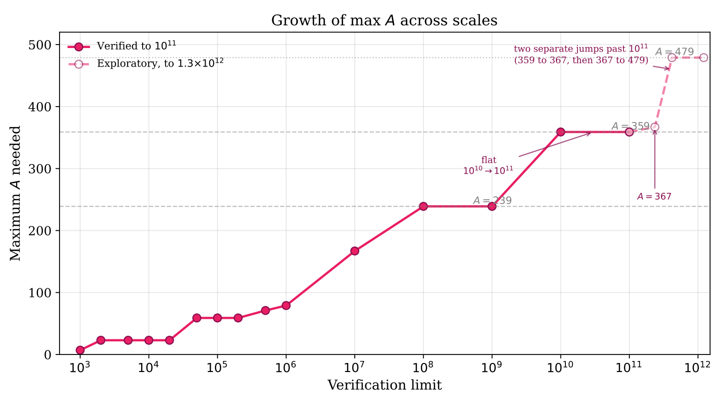

FIGURE 5. Maximum gateway parameter A required as the verification limit increases from 10^3 to 10^{11} (solid) and, from an exploratory search at the same standard described in the remark above, to 1.3×10^{12} (dashed). The growth is not well described as "roughly logarithmic" once the extended range is included: two of the four jumps in the entire record sequence occur after 10^{11}.

7.2. **Heuristic support from Elsholtz–Tao density.** Elsholtz and Tao [3] showed that the average number of solutions satisfies

$$N \log^2 N \ll \sum_{p \le N} f(p) \ll N \log^2 N \log\log N,$$

so the average number of solutions per prime up to N is $\Theta((\log N)^3)$.

For a fixed prime A, the number of divisors of $N^2 = \left(\frac{p+A}{4}\right)^2$ is $\tau(N^2) = \prod_i (2e_i + 1)$ where $N = \prod p_i^{e_i}$. On average, $\tau(N^2)$ grows like $(\log N)^c$ for a constant $c > 0$. Under a uniform distribution heuristic, the probability that none of these divisors falls in the target residue class $-B \pmod A$ is approximately

$$\left(1 - \frac{1}{A}\right)^{\tau(N^2)} \to 0 \quad \text{as } N \to \infty.$$

This suggests that for any fixed A, the set of primes for which A fails has density zero—consistent with Tao's "almost all" result. That conclusion is no longer only heuristic: Theorem 5.6, applied to the singleton set $S = \{A\}$, proves it unconditionally with the rate $O(X(\log X)^{-3/2})$, and the same theorem handles simultaneous failure across any fixed finite set of gateways, at the rate $O(X(\log X)^{-1-|S|/2})$. What the density argument does not deliver—and what Theorem 5.6 equally does not, since its constant is not uniform in T—is any statement about the 31 distinct gateway values being enough for *every* prime, which is Conjecture 7.1.

The Elsholtz–Tao bounds explain why small A dominates: with $\tau(N^2) \sim (\log p)^c$ divisors and a target residue class of density $1/A$, the expected number of valid divisors is $\sim \tau(N^2)/A$, which is larger for smaller A. The observed distribution—68.9% at $A = 7$, 91.0% by $A = 11$—is consistent with this heuristic.

7.3. **Comparison with the Salez modular sieve.** The Salez algorithm [9], extended by Mihnea and Bogdan [7] to 10^{18}, takes a fundamentally different approach. It constructs modular filters S_m—sets of residue classes modulo m for which the conjecture is provably true—and only checks primes outside these filters. At filter level R_8 (modulus $G_8 \approx 2.6 \times 10^{10}$), only ≈ 2.1 million residue classes of G_8 require direct verification, representing $\approx 0.008\%$ of all primes.

	Salez sieve	Gateway decomposition
Question	Existence	Structure
Method	Modular filter + spot-check	Classification by A
Output per prime	Yes/no	Explicit (x, y, z) with A
Verification record	10^{18}	10^{11}
Reveals A-distribution	No	Yes

The two approaches are complementary. The Salez sieve reaches much higher limits by avoiding per-prime algebraic work; our approach produces the structural data underlying the bounded-A conjecture.

7.4. **The quadratic reciprocity obstruction.** Tao [14] observed that quadratic reciprocity blocks any system of covering congruences from resolving the Mordell subset: the residue classes $\{1, 121, 169, 289, 361, 529\}$ modulo 840 involve perfect-square residues that cannot be eliminated by further modular identities. Extending the mod-840 coverage to a complete covering system therefore has a fundamental ceiling.

The gateway decomposition circumvents this obstruction entirely. Rather than seeking a congruence condition that *all* primes in a residue class satisfy, it finds a divisor of N^2 in a specific residue class—a condition that depends on the *arithmetic* of each individual prime, not on its residue class alone. This is why the gateway can (empirically) handle cases that covering systems cannot.

7.5. **The precise open problem.** Since every prime outside the residual class is covered by Propositions 4.1–4.3, the Erdős–Straus conjecture reduces to:

> For every Case B QR7 prime p, there exists a prime $A \equiv 3 \pmod 4$ and a divisor d of $\left(\frac{p+A}{4}\right)^2$ satisfying $d \equiv -p \cdot \frac{p+A}{4} \pmod A$.

This is a question about the *equidistribution of divisors in residue classes* for integers of the form $\left(\frac{p+A}{4}\right)^2$, a topic connected to work of Hooley and Tenenbaum on divisor distribution. The bounded-A conjecture strengthens this to ask whether A can be taken from a *fixed finite set*, independent of p.

7.6. **Connection to other recent work.** Ghermoul [4] constructs four explicit polynomial families in three variables that conjecturally cover all integers $\equiv 1 \pmod 4$, with one family alone covering all primes to 1.2×10^{10}. This is related to our framework: each A defines a polynomial family, and our observation that 32 values of A suffice to 10^{11} is a quantitative version of his coverage conjecture.

Our gateway decomposition produces predominantly Type II solutions in the Elsholtz–Tao classification (where p divides two of the three denominators). The structural reason is that y and z in Theorem 3.1 satisfy $y = (B + d)/A$ and $z = By/d$ with $B = pN$, so $p \mid z$ automatically whenever $p \nmid d$.

8. Conclusion

We have presented a structural analysis of the Erdős–Straus conjecture through a gateway decomposition parameterised by a single integer A. Using 32 prime values of A (all at most 359) for the primes $p \equiv 1 \pmod 4$, together with the classical $A = 1$ decomposition for $p \equiv 3 \pmod 4$, we verify that every prime $p \leq 10^{11}$ admits an explicit decomposition $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$.

Section 5 proves unconditionally that the residual class left to that search is negligible: it has relative density $O((\log X)^{-1/2})$ among the primes, and the first gateway $A = 3$ already succeeds for a set of primes of density one—the best possible statement of that kind, since $A(p) \geq 3$ always. The proof is an elementary Selberg sieve on the integers and uses no equidistribution theorem. It also corrects an argument given in an earlier version of this paper, which drew the same conclusion from a Landau-type count of integers that does not support it (Remark 5.7).

The main computational observation is the slow growth of the gateway parameter: $\max A$ increases in occasional large jumps separated by long flat stretches (79 at 10^6, 239 at 10^8, 359 at 10^{10}, holding through 10^{11}). A flat stretch of this kind is not, by itself, evidence of an eventual ceiling: Section 7 shows two models fitting the verified data identically, one bounded and one not, and an exploratory search past 10^{11} has already found two further jumps, to 367 and 479. If $\max A$ is absolutely bounded (Conjecture 7.1), the full conjecture follows.

The framework provides a potential bridge between computational verification and a theoretical proof: the bounded-A conjecture is a concrete statement about divisor equidistribution in residue classes for structured quadratic integers, a topic accessible to the methods of analytic number theory (Remark 5.11 locates the nearest published results).

Code availability. All verification code (Python) and reproducibility data are available at `https://github.com/m4cd4r4/erdos-straus-gateway`.

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